Degree Program Mechatronics/Robotics (Master)
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DEVELOPMENT OF DUST AND MILLING CHIPS EXHAUST SYSTEM FOR ROBOT MACHINING STATIONS

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Abstract: *Industrial robots present a distinct advantage in performing tasks such as milling, cutting, and polishing on large workpieces, as the workpiece does not have to be attached to a certain machine table or jaw chuck as required in traditional subtractive manufacturing methods. With the increasing demand on industrial robots for machining, many leading companies in industrial robotics, such as ABB, KUKA and Fanuc, have integrated subtractive manufacturing processes in their robot programming software (e.g. PowerPac in ABB RobotStudio) in order to facilitate generating the robot code for these processes.*
Although a flexible working place is principally advantageous, the flexible poses of robotic arms for different operations poses a challenge for designing and positioning exhaust systems to extract the dust particles emitted during the machining process, which can threaten employee health. A dust extractor must be placed as closely as possible to the cutting/milling tool without geometrically colliding with the robot task, which clearly becomes a problem if the robot pose is constantly changed. The two most common solutions are to either enclose the robot in a chamber that circulates and purifies the surrounding air, thus keeping the contaminated air inside, or to mount the suction cup far from the end effector in order to avoid collision, and increase the suction power in order to extract the dust particles. The latter solution is clearly more flexible, widely applicable and cost-effective. However, positioning the extractor and suction power at a standard setting for all operations increases energy usage and costs. The main challenge is to create a dust extractor that automatically moves the dust inlet closer to the workpiece without colliding with its surroundings.
This study describes the development of a dust extractor for the UAS Technikum-Wien Digital Factory milling/cutting/polishing robot that controls the distance of the suction head to the end effector according to the length of the used tool, which leads to reducing the necessary suction power. This tool is provided with sensors that measure the distance to its surroundings in order to avoid collisions dynamically, and thus eliminating the need for the robot programmer to consider the dust extractor during the programming process.
Keywords: robot machining, exhaust station, milling chips, dust, automatic tool changer.

1. INTRODUCTION

The future manufacturing is characterised by high customisation. Currently, CNC machine tool are widely used for machining operations, since they are able to deliver higher machining accuracy with high stability; however, simultaneously, their costs are high and their functions are single. Therefore, a multi-function and low-cost machine is a trend to replace the current machine tools, e.g. industrial robot could be a potential one. There are many important factors in robotic machining dynamics that are crucial for high quality machining, including robot stiffness and machining vibration [1].
An important by-product of subtractive manufacturing (Milling, cutting, etc.) is the creation of dust and milling chips. The type of produces chips and produces dust depend on the machined material, machining tool and speed of cutting.
While the concern of extracting the milling chips is centred on maintaining a clean workspace, the dust created by the machining process can pose a threat to the health of people surrounding the workstation.
There are many assessment methods for identifying the particle size of created dust. One of them is laser diffraction method using a laser particle sizer. A particle-size distribution of the collected sample can be generating using this method [2].

The induced fumes in gas metal arc welding (GMAW) and flux cored arc welding are also subject to emission limits to ensure workers health. A recent NSRP study determined lost productivity and other costs of complying to proposed limits for hexavalent chromium at Navy and private shipyards could cost millions of dollars annually [3].

This study aims at developing an exhaust station for the KUKA KR-16 robotic arm station in Figure 1. The exhaust station shown in the figure has been used shortly then omitted due to ineffectiveness. This study will aim at creating an exhaust inlet with a software-controllable length (Figure 2), so that it can retract when the robot is changing tools then extend to the needed length after the tool has been gripped. Shortening the distance to the milling area will enable the exhaust station to produce a stronger vacuum at the area where the dust and milling chips are created, and thus eliminate more of them than the existing station.

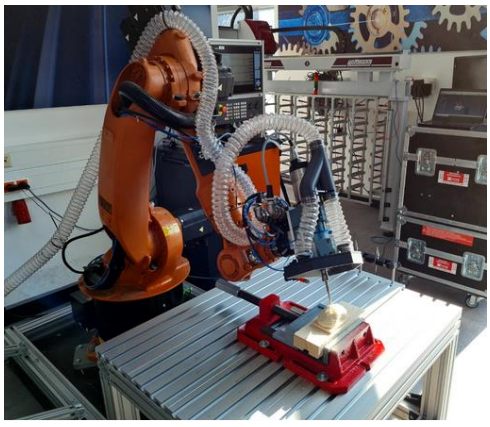


Figure 1: KUKA KR-16 station in the Digital Factory - Technikum Wien

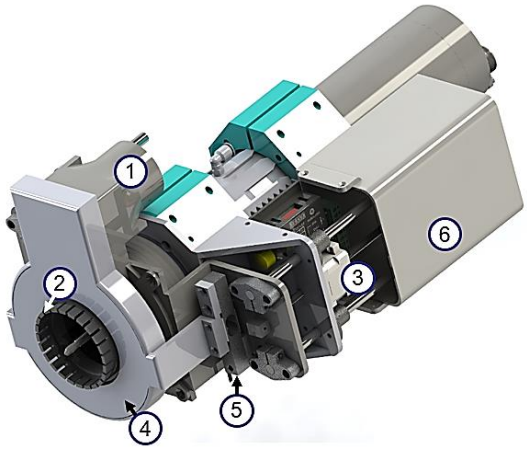


Figure 2: Developed exhaust intake with variable length.

1: Exhaust tube inlet – 2: Flexible Brush (black) – 3: Non-captive linear actuator – 4: Collision protector – 5: Load cell – 6:Covering lid for electronics

2. STATE OF THE ART

Many methods have been used for extracting dust and milling chips. These methods deliver different results in terms of flexibility, energy consume and their effectiveness in extracting dust, fumes, and milling chips.

2.1. Hooded exhaust systems

Extraction hoods are primarily used in applications that include a higher risk of exposure to poisonous gases, such as arc welding. They are also effective in extracting dust created by milling or polishing [4].
For a better performance, special fins can be hanged from the cap to create an enclosure around the working area as showed in Figure 3.



Figure 3: Exhaust hood for arc welding developed - Valco Entstaubungstechnik [4]

2.2. Dust shoes

Dust shoes are mainly used in 3-axis CNC milling machines. Their long brushes enable them to cover the area around the milling head, reaching down to the milling table, thus creating a good vacuum for sucking dust and milling chips.



Figure 4: Dust shoe with flexible brushes - Avid CNC [5]

This method is hard to use in robotic machining station, as its rigidity will pose a challenge to programming a trajectory path that does not cause a collision between the dust shoe and the workpiece or the fixture tools.

2.3. Downdraft tables

Downdraft tables are common in woodwork workshops, as they combine the functionality of both a working table and an exhaust station. They can be connected to an external vacuum station to extract the milling chips [6].



Figure 5: Downdraft table with an external plug to a vacuum station - GEMAtech [6]

2.4. Bendable robotic fume extractors

Exhaust stations with bendable designs have been developed in order not to compromise the flexibility of robotic arms. An example of those designs is xFUME™ ROBO Robotic Fume Extraction from Abicor Binzel, which fits over the front module of the robotic welding torch to capture weld fumes at the welding arc [7].



xFUME™ ROBO Robotic Fume Extraction [7]

3. RESEARCH PROBLEMS AND OBJECTIVES

Maintaining the flexibility of a robotic arm is a key research goal in this study, therefore, the designed dust exhaust should fulfil many requirements in its geometry, efficiency and programming method. The following points are problem that can be created when adding a dust exhaust to a robotic arm.

3.1. Interfering with the tool change

A system for automatic tool changing has been developed for Kuka KR-16 station, which is based upon a fork-like holder for different chuck collets. The tool changing process consists of the robot receiving the command with the slot upon which the required tool is mounted, it goes to the right position in a predefined path, then activated the collet clamp. In order for this procedure to take place correctly, the suction cup of the exhaust station shouldn't exceed the height specified in Figure 6, as it is necessary in order for the form to hold the tool correctly.

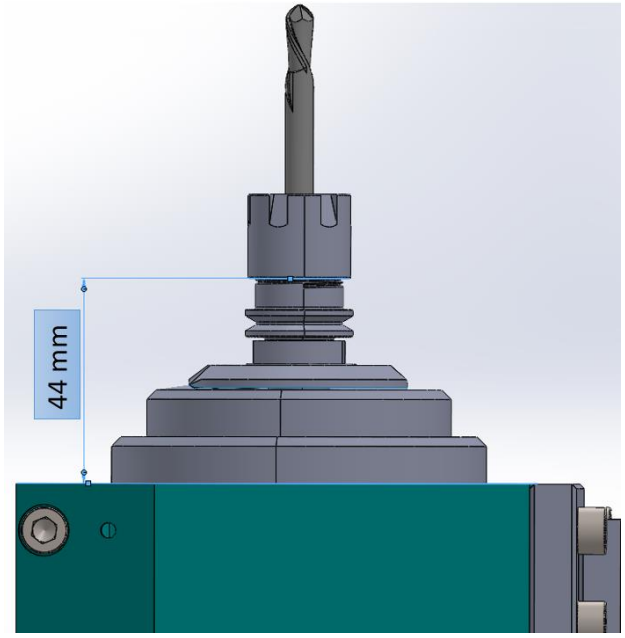


Figure 6: Available distance for the exhaust station

3.2. Variable length of the machining tool

The efficiency of the exhaust system increased as the distance between the dust and the suction intake decreased. It is therefore crucial for the exhaust system to vary its length according to the used tool, in order not to compromise its efficiency or limit the used to a certain tool size.

3.3. Static obstacle avoidance

Obstacle avoidance is an important point to consider when creating the robot code for a specific tast. Many software, including RoboDK, have a tool for checking collisions along the path. An important thing to consider when designing the dust exhaust is to ensure that its geometry doesn't compromise the reach of the robotic arm. Figure 7 shows a collision between the spindle and the fixture plate could limit how far the milling tool can reach into the workpiece.

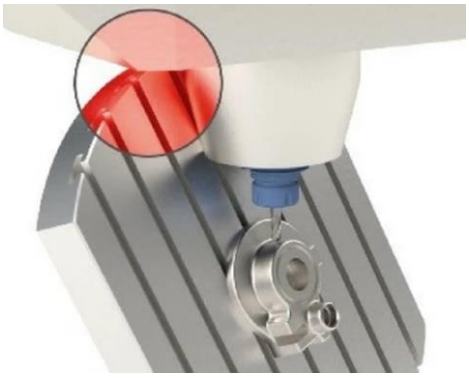


Figure 7: Collision of the robotic arm with the fixture plate

3.4. Dynamic obstacle avoidance

In addition to the proactive approach in obstacle avoidance when planning the robot path, dynamic obstacle avoidance can detect a collision with an object that is unaccounted for in the path planning and control the robot path accordingly. This is important to the function of an exhaust system that aims to move the exhaust intake as close as possible to the end effector. An ideal scenario for drilling deep holes inside an object would be for the exhaust intake to expand and come close to the end effector, then retract its length accordingly as the end effector drills deeply into the workpiece in order to avoid collision. This approach requires a sensor technology that detects the distance between the exhaust intake and any possible obstacles. The distance information given by the sensor system should then be used to either change the position of the exhaust intake, or to stop the robot motion in case the retract of the intake is not enough to avoid collision.

3.5. Integrating the exhaust intake position into the robot code

Generating the robot code for a specific task includes many steps, starting from the CAD design, then generating the end effector path with the desired tool, and ending with finding the right robot configuration for the needed end effector path. The first two steps are carried out in Solidworks and its CAM extension, while the last one is programmed in RoboDK. Since the variable tool length criteria requires the exhaust intake to change its position according to the tool size, this poses the need for the information to be embedded into the robot code and sent to the exhaust system at the right time. Many robot controls, including the Kuka KRC4, contain peripheral control units that can be used to control different types of accessories around the robot station.

4. MATERIALS AND METHODS

In this section, we will go into details of the design of the dust extractor and reasons for choosing the main components such as the actuator, force sensor and microcontroller. Figure 8 shows the air flow through the dust extractor from the inlet with flexible brushes to the tube connector.

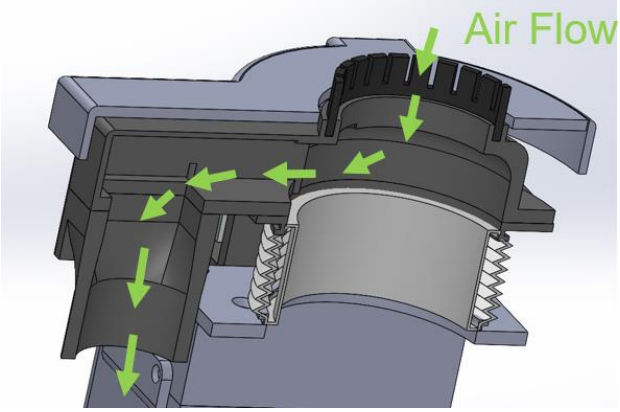


Figure 8: Section view: air flow through the dust extractor

4.1. Actuator: Non captive motor

None-captive stepper motors (Figure 9) are characterized by the lead screw going through the motor core and being rotated when the motor moves. If the lead screw is blocked from rotating and the motor is mounted on a linear guide, rotary movement can be converted into linear movement [8].

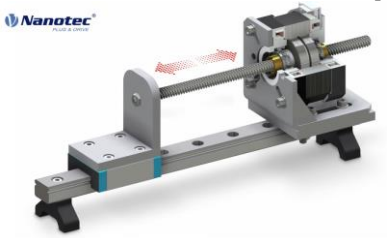


Figure 9: Non-captive linear actuator

Source: Nanotech Electronic GmbH

This method has the advantage of not needing a brake (e.g. electromagnetic brake) to block the movement of the linear actuator when it is powered off.

4.2. Collision sensor: load cell

The most commonly used type of load cell in industrial applications is the strain gauge load cell. A strain gauge load cell is comprised of a solid metal body (or "spring element") on which strain gauges have been secured. When a load is applied, the body of the load cell is slightly deformed, but, unless overloaded, always returns to its original shape. In response to the body shape changes, the strain gauges also change shape. This, in turn, causes a change in the electrical resistance of the strain gauge which can then be measured as a voltage change. Since this change in output is proportional to the amount of weight applied, the weight of the object can then be determined from the change in voltage **[9]**.

$$V_0 = \left[\frac{R_3}{R_3 + R_4} - \frac{R_2}{R_1 + R_2}\right] . V_{EX} \qquad (1)$$

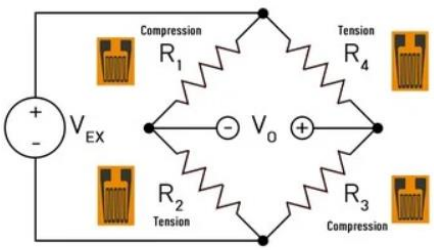


Figure 10: Wheatstone measurement bridge

Source: 800loadcel.com

A collision protector has been mounted on the loadcells to detect a collision before it reached the main body of the load extractor, and thus sending a signal for the motor controller to drive back.

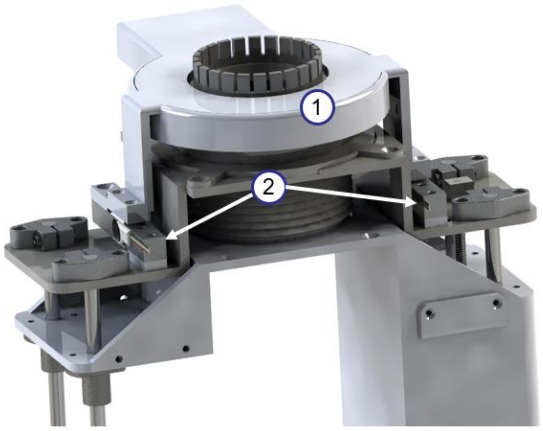


Figure 11: Collision detector mounted on two load cells
1: Collision protector – 2: Load cells

4.3. Sequence of operation diagram

The dust extractor is controlled by the KUKA KR C4 controller using an analog output signal, which is received by the microcontroller (NodeMCU) that controls the stepper motor controller (Figure 12). The set point (position) of the dust extractor corresponds to the analog signal via a predefined voltage to length conversion ratio.

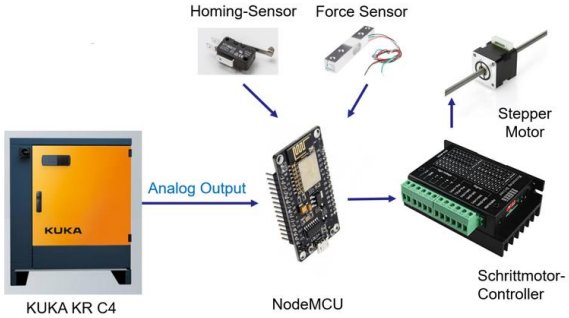


Figure 12: Dust inlet position control system

The operation sequence (Figure 13) starts by powering on the circuit, followed by going to the home position. Upon that, the set point defined by the analog signal from KUKA KR C4 is read and the microcontroller tries to drive to that point while checking that the load cell doesn't indicate a collision. If a collision is detected while the robot is at the zero position (driving backwards not possible), an Alarm signal is sent back to the robot controller.

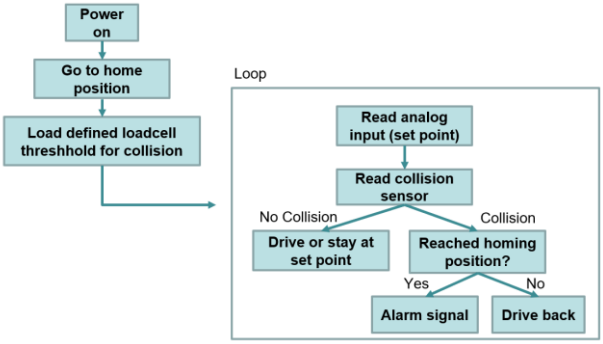


Figure 13: Position control diagram

5. RESULTS

5.1. 3D printing the dust extractor

The design is fabricated from PLA using an FFF 3d printer. Threaded inserts have been used to mount many components, including the cylinder rail guides and holders, load cells and stepper motor. The whole design requires approximately 900g of PLA.

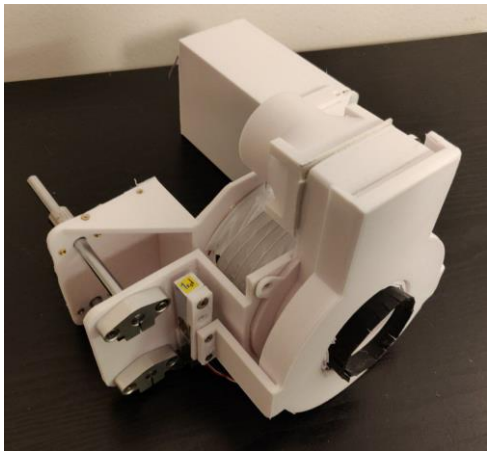


Figure 14: 3D printed design

Although most components were stable enough for the dust extractor to function properly, the movable part of the design was wackling slightly due to its bendability, which causes the left part in Figure 15 to follow the movement of the right part (connected directly to the lead screw of the linear actuator) with a slight delay.
It is recommended increase the corner supports of this part in order to decrease its bendability, or to print it from a stiffer material, such as carbon fibre.

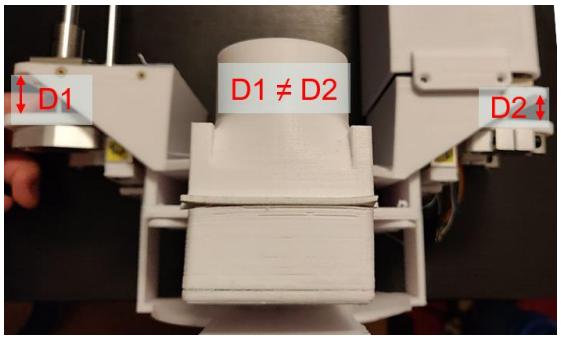


Figure 15: Mismatch in the position of the left side and right side of the movable dust inlet.

The load cell is fixed upon the load cell using an L-shaped adapter. It is recommended to manufacture this adapter from another material, as it is expected to bend over time, since it carries the force of every collision that the collision protector receives. A bend in this adapter over time will require a new calibration of load-cell threshold values in the software.

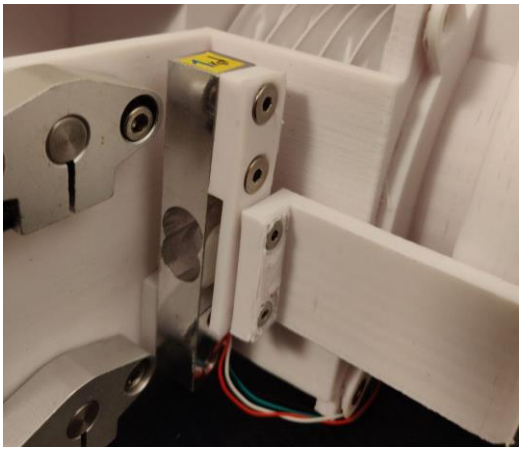


Figure 16: High loads on the collision protector holder can cause it to bend

5.2. Extracting milling chips while milling Styrofoam

The dust extractor was tested by mounting it on a cartesian CNC milling machine, then running a simple milling program on a Styrofoam piece in order to measure the extractors effectiveness in collecting dust particles and millings chips.

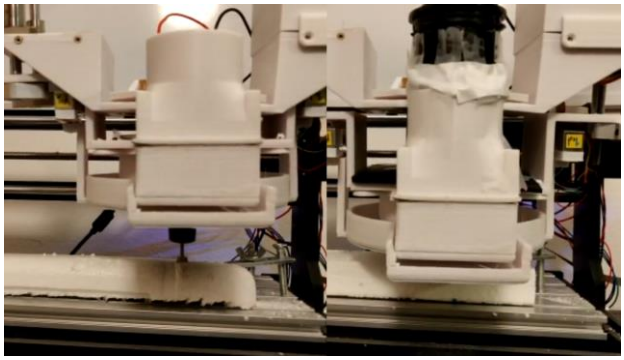


Figure 17: Mounting the dust extractor on a cartesian CNC milling machine. Dust extractor retracted (left) then extended (right).

The dust extractor could collect both small and big particles using a 200W vacuum cleaner.

5.3. Collision detection

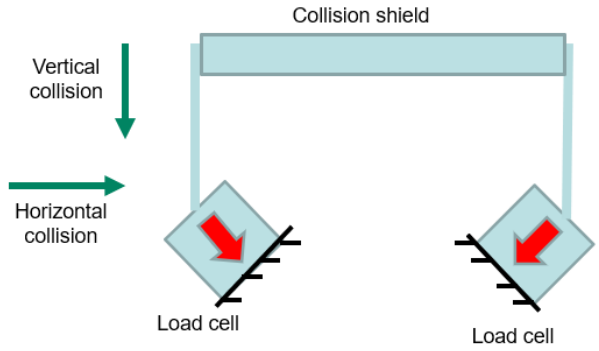
The following approach was used to detect collisions: Two threshold values were defined for weak and strong collisions accordingly. When a strong collision is detected, the dust inlet retracts quickly, while it stays in its current position if a weak collision is detected. Figure 18 Shows how the dust inlet stops before reaching the position setpoint when it confronts an obstacle.



Figure 18: Dust extractor detecting an obstacle before reaching the setpoint

6. OUTLOOK ON FURTHER DEVELOPMENT

Although the collision protector in its current state can detect very small vertical collisions that press the collision protector along its movement axis, it is insensitive to horizontal collisions, as the load cell is only sensitive in one direction. It is recommended to mount the load cells in 45° orientation in order to detect a wider spectrum of collisions.



7. CONCLUSION

A motorized dust extractor that avoids collisions with its environment is a very effective approach for extracting dust particles and milling chips, as it can move closely to the milling bit and thus requiring a low-power vacuum cleaner in comparison to the conventional methods in dust extraction. Force sensors can be used to dynamically avoid collisions when designed with a suitable collision protector that detects a collision before it reaches the body of the dust extractor, giving the linear actuator enough time to retract to a safe position.

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